RMG Manufacturing & Business Performance Dashboard

Data Profiling & Preparation Report

Version: Final

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1. Data Overview

The project uses 13 datasets covering key areas of RMG business and manufacturing operations.

The datasets are divided into **5-dimension tables** and **8 fact tables**. Together, they provide information about buyers, factories, production lines, styles, dates, orders, shipments, production, cutting, knitting, dyeing, fabric inspection, and finishing.

The data was structured and prepared for analysis in PostgreSQL and Power BI.

2. Dataset Summary

Field	Details
Project Name	RMG Manufacturing & Business Performance Dashboard
Document	Data Profiling & Preparation Report
Industry	RMG Sector
Tools	Excel, PostgreSQL
Source	Kaggle
File Format	CSV
Number of Files	13

Dimension Tables

Table	**Purpose**
dim_buyer	Buyer information
dim_date	Date and calendar information
dim_factory	Factory information
dim_line	Production line information
dim_style	Product/style information

Fact Tables

Table	**Purpose**
fact_order	Customer order information
fact_shipment	Shipment and freight information

Table	Purpose
fact_production_daily	Daily production performance
fact_cutting_daily	Daily cutting performance
fact_knitting	Knitting machine performance
fact_dyeing	Dyeing batch performance
fact_fabric_inspection	Fabric inspection results
fact_finishing_daily	Daily finishing and packing performance

3. Fact & Dimension Design

The data model follows a fact and dimension structure to organize business transactions and support descriptive information.

Dimension Tables

Dimension tables provide descriptive information that is used to filter, group, and analyze business activities.

- dim_buyer — Buyer information
- dim_date — Date and calendar information
- dim_factory — Factory information
- dim_line — Production line information
- dim_style — Product and style information

Fact Tables

Fact tables contain measurable business and operational activities.

- fact_order — Customer order information
- fact_shipment — Shipment and freight information
- fact_production_daily — Daily production performance
- fact_cutting_daily — Daily cutting performance
- fact_knitting — Knitting machine performance
- fact_dyeing — Dyeing batch performance
- fact_fabric_inspection — Fabric inspection results
- fact_finishing_daily — Daily finishing and packing performance

This structure separates descriptive attributes from measurable business activities and supports consistent analysis across time, buyers, factories, production lines, and styles.

4. Data Model & Relationships

The main relationships are based on shared keys between fact and dimension tables.

Examples include:

- dim_buyer[BuyerID] → fact_order[BuyerID]
- dim_factory[FactoryID] → relevant fact tables
- dim_date[DateKey] → date-related fact tables
- dim_line[LineID] → fact_production_daily[LineID]

- dim_style[StyleID] → relevant order, production, cutting, and finishing tables

The model enables consistent filtering and aggregation across the Power BI dashboard.

5. Data Granularity

Each fact table represents a different level of business activity:

Fact Table	Granularity
fact_order	One row per order
fact_shipment	One row per shipment
fact_production_daily	Order + Factory + Line + Style + Date
fact_cutting_daily	Order + Factory + Style + Date
fact_knitting	Machine + Date
fact_dyeing	One row per dyeing batch
fact_fabric_inspection	One row per inspection
fact_finishing_daily	Order + Factory + Style + Date

Understanding the granularity helps ensure that KPIs and aggregations are calculated correctly.

6. Data Cleaning & Preparation

The datasets were reviewed and prepared before being used for analysis.
The preparation process included:

- Reviewing table structures and data types
- Checking data consistency
- Validating key fields and relationships
- Preparing date fields for time-based analysis
- Reviewing numerical and percentage fields
- Preparing the data model for Power BI
- Preparing data for KPI and dashboard analysis

7. Data Quality Considerations

The analysis considers common data quality areas such as:

- Missing or invalid values were reviewed.
- Duplicate records were checked.
- Data types were reviewed for consistency.
- Key consistency between related tables was validated.
- Numerical and percentage fields were reviewed.
- Categorical values were checked for consistency.

These checks help maintain reliable calculations and dashboard results.

8. DAX & KPI Preparation

DAX measures were created in Power BI to calculate the main business and operational KPIs.

The measures cover:

- Order value and order performance
- Shipment performance
- Production achievement and efficiency
- Lost time
- Finishing rejects
- Dyeing batch loss
- Dyeing reprocessing
- Finishing packing performance

The measures are designed to respond dynamically to the dashboard filters.

9. Data Preparation Workflow

The overall data preparation and reporting workflow is:

Raw CSV Data → Data Inspection → PostgreSQL Analysis → Data Validation → Power BI Data Model → DAX Measures → Dashboard

This workflow ensures that the final dashboard is based on structured data and consistent calculations.

10. Conclusion

The datasets were profiled and prepared for analysis. Key data quality areas were reviewed, and the data was considered suitable for SQL analysis, data modeling, and Power BI dashboard development.